

MASTER PROJECT

Shock propagation through the lens of remittances

Who bears the cost? Evidence from US–Mexico migration corridors

Authors

Anqi Wang Liu, Francesco Zucca, Joan Costa Rodríguez,
Michael Fehl, Xènia Alemany Rovira

Barcelona School of Economics

Master's program in Economics

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Abstract

1. Introduction

In 2024, Mexico received more than 67 billion current US dollars in remittances, making it the second-largest recipient country in the world after India (World Bank, 2024). This figure represents approximately 3.7% of Mexico's GDP (reference?), highlighting the macroeconomic significance of these transfers. With roughly 96% of remittances to Mexico originating from the United States (reference?), and the average Mexican household in the US sending approximately US\$12,800 per year (Isidro Luna & Martínez Hernández, 2026), a natural question arises: does an economic shock in the US propagate to Mexico through the remittance channel? Specifically, do Mexican migrants absorb the shock and maintain their transfer levels constant, or do they reduce remittance outflows in response to a decline in their current income or wealth? The answer has direct implications for understanding the vulnerability of Mexican households to external shocks and the extent to which the remittance channel acts as a shock transmission mechanism.

The closest paper to ours is Caballero et al. (2023), who study how the Great Recession affected remittance receipt among Mexican households. Using a shift-share research design that exploits variation in migration networks across Mexican municipalities, they find that households more connected to severely hit U.S. labor markets experiencing larger employment declines were significantly less likely to receive remittances. This confirms that the remittance channel acts as a transmission mechanism for US labor demand shocks. However, their analysis is limited to the extensive margin (i.e. whether the households receive remittances at all), while not providing any estimate for the intensive margin (i.e. how much remittance flows decline in response to the shock). Quantifying this magnitude is central to our paper, as it determines the actual income loss borne by Mexican households. Moreover, the Great Recession was a global shock of exceptional magnitude, which simultaneously contracted both the U.S. and Mexican economies, with Mexico recording a 6.6 points decline in real GDP in 2009 (Villarreal, 2010). This makes it difficult to isolate the effect of the U.S. economic decline on Mexican remittance inflows. Furthermore, shocks of this magnitude are uncommon, which limits the external validity of the findings derived from them.

We address these challenges by exploiting the localized and more frequent nature of natural

disasters. From 1980 to 2024, the U.S. has sustained 403 weather and climate disasters for which the individual damage costs reached or exceeded US\$1 billion, with cumulative costs that exceeds US\$2.915 trillion (NOAA National Centers for Environmental Information, 2025). These events generate localized reductions in firm productivity, leading to lower family income (Boustan et al., 2020) and therefore plausibly reducing the capacity of affected migrant households to remit.

The relationship between natural disasters and remittances has been studied before, but existing work focuses on the opposite side of the corridor, analyzing what happens to bilateral remittance flows when the *receiving* country is struck. Bettin et al. (2023) study the evolution of remittance flows from 103 Italian provinces to 79 developing countries and find that remittances increase by approximately 1.8% on impact following a disaster in the recipient country, rising to about 2.4% four months later and rapidly decreasing after one year. This provides a useful benchmark for the speed and persistence of remittance adjustment, although it operates through a different mechanism than ours, where the shock originates on the *sender* side.

We contribute to this literature by providing causal evidence on how a localized US shock propagates to Mexico through the remittance channel. To this end, we exploit Hurricane Ian (September 2022), the costliest hurricane in Florida's history, which caused US\$ 109.5 billion in damage (Bucci et al., 2026), as a natural experiment. The storm struck the southwestern coast of Florida but did not affect Mexico, providing a clean one-sided shock to the sender side of the remittance corridor. We estimate bilateral remittance flows between each US state and each Mexican municipality by applying a migration weighting matrix to Banxico data on remittance inflows by U.S. State of origin (Banco de México, CE168, 2026). Then, we implement a difference-in-differences design comparing Florida-linked municipalities against those connected to unaffected states. We find that ... Our findings suggest that ..., implying that ...

2. Theoretical framework and empirical strategy

This chapter outlines the theoretical and empirical framework use for our analysis. We begin by adapting a heterogeneous-firm trade model to the context of individual remittance behavior, establishing the theoretical foundations of how a migrant's transfer decisions can be influenced by economic shocks. Next, we introduce Hurricane Ian, which hit Florida in 2022, as a natural experiment that acts as an exogenous shock to migrant income. Finally, we present our estimation strategy (explain the strategy here) and discuss how we examine potential spillover effects, which can bias our estimates if not properly accounted for.

2.1. Remittance model

To analyze the impact of shocks on remittance flows, we adapt the logic of heterogeneous-firm trade models to the context of remittances. Specifically, we consider a model where migrants differ in income and altruism intensity, and where remitting requires paying both a fixed access cost and a variable cost proportional to the amount remitted. Formally, consider migrant i from origin municipality m in year t . The migrant faces a utility maximization problem under a CES aggregator over local consumption and total resources available to the origin household:

$$U_{imt} = \left[\alpha \left(c_{imt}^L \right)^{\frac{\sigma-1}{\sigma}} + (1 - \alpha) \left(\phi_{im} \left(R_{imt} + y_{mt}^O \right) \right)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \quad (2.1)$$

where $\phi_{im} > 0$ denotes the altruism or obligation intensity parameter, analogous to the productivity parameter in Melitz (2003). This strictly positive nature of this parameter ensures that sending remittances always increases the migrant's utility. The variable $c_{imt}^L \geq 0$ represents the migrant's consumption in the destination country, $R_{imt} \geq 0$ is the amount sent in remittances, and $y_{mt}^O \geq 0$ captures the origin household's income as observed by the migrant. The weight of local consumption is given by $\alpha \in (0, 1)$, and the elasticity of substitution between local consumption and origin-household resources is given by σ . Notably, origin-household total resources are defined as $R_{imt} + y_{mt}^O$, which we treat as a single composite good. The presence of y_{mt}^O introduces a role for origin-country economic conditions: negative income shocks at origin (i.e. lower y_{mt}^O) increase the marginal utility of remittances, generating an insurance motive consistent with Yang and Choi (2007).

The migrant's budget constraint is given by:

$$c_{imt}^L + f_{mt} \mathbb{1}\{R_{imt} > 0\} + (1 + \tau_{mt})R_{imt} \leq w_{imt} \quad (2.2)$$

where $w_{imt} \geq 0$ is the migrant's wage, $f_{mt} \geq 0$ is a fixed remitting cost capturing the per-transaction expense of accessing a remittance channel, analogous to the fixed export cost in Melitz (2003), and $\tau_{mt} > -1$ is the variable cost which scales with the amount remitted, analogous to the iceberg cost structure in Melitz (2003). Notice that, even though τ_{mt} is typically positive to reflect transaction fees or exchange rate spreads, we allow for the possibility of negative variable costs, which could arise from the presence of remittance subsidies or favorable exchange rate regimes.

Assuming that the migrant maximizes utility subject to the budget constraint and chooses to remit a positive amount ($R_{imt} > 0$), solving the Lagrangian yields the optimal remittance function:

$$R_{imt}^* = \frac{(w_{imt} - f_{mt} + (1 + \tau_{mt})y_{mt}^O) A_{imt} - y_{mt}^O}{1 + (1 + \tau_{mt})A_{imt}} \quad (2.3)$$

where we define the parameter A_{imt} as:

$$A_{imt} \equiv \left(\frac{(1 - \alpha)\phi_{im}}{\alpha(1 + \tau_{mt})} \right)^\sigma \quad (2.4)$$

The term A_{imt} represents the optimal ratio of origin-household resources to own local consumption, adjusted for the variable cost of remitting. It comprises the migrant's altruism intensity (ϕ_{im}), the utility weight of own consumption (α), the elasticity of substitution (σ), and the variable cost of remitting (τ_{mt}).

To understand the effect of destination-country economic shocks on remittance flows, we analyze changes in the migrant's wage (w_{imt}). The partial derivative of optimal remittances with respect to wage is strictly positive:

$$\frac{\partial R_{imt}^*}{\partial w_{imt}} = \frac{A_{imt}}{1 + (1 + \tau_{mt})A_{imt}} \quad (2.5)$$

Therefore, through an income effect, the model predicts that a positive shock to the migrant's

wage will increase remittances, while a negative shock will decrease them. The fixed cost f_{mt} establishes an entry threshold: if the migrant's wage is not sufficiently high relative to the fixed cost, they will not send remittances at all. Additionally, the variable cost τ_{mt} reduces the marginal benefit of each dollar sent, dampening the sensitivity of remittances to wage fluctuations.

Ultimately, this model provides a theoretical framework to understand how shocks that affect migrants' income can influence remittance flows, which is the main focus of our empirical analysis. However, a negative shock to migrant's wage does not directly imply a significant decrease in remittances, as the response depends on the shock's magnitude, the migrant's altruism intensity, and the cost structure of remitting.

2.2. The natural experiment: Hurricane Ian (2022)

The shock selected for this analysis is Hurricane Ian, which struck the southwestern coast of Florida on September 28, 2022, with peak sustained winds of 140 kt (around 260 kph) (Bucci et al., 2026). Ian caused an estimated US\$ 112.9 billion in total damage in the United States, of which US\$ 109.5 billion occurred in Florida alone, making it simultaneously the costliest hurricane in Florida's history and the third-costliest in US history. In Lee County alone, at least 52,514 structures were impacted, of which 5,369 were completely destroyed (Bucci et al., 2026).

Crucially for our identification strategy, Hurricane Ian did not affect Mexico. The storm originated in the Caribbean, made landfall in Cuba and Florida, and dissipated over the Atlantic after a final less impactful landfall in South Carolina (Bucci et al., 2026). Therefore, the shock is spatially clean. Indeed, remittance senders in Florida were directly exposed to a severe and sudden wealth shock (reference or data?), while remittance receivers in Mexican municipalities were entirely unaffected. This allows us to better isolate the transmission of the shock through the remittance channel.

2.3. Estimation technique

Nothing works we are doomed

2.4. Examining spillover effects

A fundamental requirement for standard causal inference models is the Stable Unit Treatment Value Assumption (SUTVA), formalized by Rubin (1980). SUTVA assumes that the potential outcomes of any unit depend exclusively on its own treatment assignment, excluding the possibility of any indirect spillover between units.

In our context, U.S. states not directly exposed to a local shock might still be affected if control units are exposed to the treatment through indirect channels, leading to biased estimates (Clarke, 2017). For instance, given that California and Texas all host a big proportion of Mexican migrants, we hypothesize later in the paper that these two states might have played a role in offsetting the negative Florida shock. If a Mexican municipality highly dependent on Florida remittances experiences an income shock due to Hurricane Ian, connected migrants residing in Texas might respond by increasing their transfers to compensate for the drop. If this was the case, **the estimated Florida-exposure effect on total remittances would capture the net effect of the shock after within-network adjustment, rather than the direct fall in Florida-origin remittance.**

Therefore, in later sections we will try to argue that our estimates are subject to general equilibrium effects. We base our analysis on the exposure-mapping logic of (Aronow & Samii, 2017) to define indirect exposure through alternative migrant networks. Rather than assuming that a municipality's remittances respond only to Florida exposure, we allow the response to vary with exposure to other major U.S. destinations, especially Texas and California.

3. Data and bilateral remittance flow estimation

This chapter outlines the data sources and the methodology we construct for estimating bilateral remittance flows between U.S. states and Mexican municipalities, which is not publicly available. First, we describe the datasets from Banco de México and Matricula Consular de Alta Seguridad (MCAS), which provide the remittance and migration data, respectively. Then, we compute a migration-share weighting matrix, which we use to distribute U.S. state-level remittance outflows to Mexican municipalities. Finally, we explore the underlying assumptions of our estimation approach, testing the persistence of migration networks over time and validating our estimates by comparing them with official data available at a more aggregate level.

3.1. Data sources

The source of remittance data is Banco de México (Banxico), which provides two complementary datasets through its Sistema de Información Económica (SIE). The first, series CE168, records remittance inflows from each US state to Mexico in millions of current US dollars at quarterly frequency, covering the period from Q1 2013 to Q1 2026 (Banco de México, CE168, 2026). This series serves as the US state outflows (r_s) for our bilateral flow estimation. The second, series CE166, records remittance inflows received by each Mexican municipality, also in millions of current US dollars at quarterly frequency and over the same sample period (Banco de México, CE166, 2026). This series acts as a benchmark to validate the accuracy of our estimates at an aggregate level.

To bridge these datasets, we utilize administrative records from the Matricula Consular de Alta Seguridad (MCAS) program, managed by the Instituto de los Mexicanos en el Exterior (IME) (Instituto de los Mexicanos en el Exterior, 2026). The publicly available version of this dataset provides a comprehensive registry of consular identification cards issued to Mexican nationals residing in the United States each year, from 2010 to 2024. While the MCAS does not constitute an exhaustive census of all migrants, it represents a robust proxy for the spatial distribution of Mexican migrants in the United States. In particular, it is highly representative of undocumented and recently arrived migrants, who face the strongest incentives to obtain consular identification (Caballero et al., 2018).

3.2. Bilateral remittance estimation

To estimate bilaterance remittance flows from U.S. states to Mexican municipalities, we construct a migration-based weighting matrix using annual MCAS from 2010 to 2024. Because these records capture new consular issuances rather than the complete stock of the remitting population, which also includes previously arrived migrants, we rely on the assumption of persistence of migration patterns over time. To mitigate the volatility of yearly migration fluctuation and construct a robust proxy for the long-term spatial distribution of Mexican migrants, we compute the average of these matrices across all years. We then apply this averaged weighting matrix to the state-level remittance outflows from Banxico's series CE168 to derive the U.S. to Mexican municipality bilateral flows.

Formally, we define $MCAS_{ms}$ as the average annual count of MCAS issuances to migrants from Mexican municipality m residing in U.S. state s . For each U.S. state, we calculate the average migration share, w_{ms} , which represents the proportion of that state s 's Mexican population originating from the m municipality, as the ratio of $MCAS_{ms}$ to the total number of MCAS issuances from that state across all municipalities M :

$$w_{ms} = \frac{MCAS_{ms}}{\sum_{m=1}^M MCAS_{ms}} \quad (3.1)$$

By applying the weights computed from equation (3.1) to the official aggregate state-level outflows from Banxico (r_s), we derive the estimated bilateral remittance flow ($\hat{r}_{ms}$) from state s to municipality m :

$$\hat{r}_{ms} = r_s \cdot w_{ms} \quad (3.2)$$

This migration share estimation approach yields the final estimate of our outcome variable.

3.3. Underlying assumptions and validation exercises

The validity of this estimation relies on two primary identification assumptions. First, we assume the temporal persistence of migration flows, implying that the distribution of migrants recorded in the 2010–2024 MCAS data is representative of the active remitting population.

Second, we assume that, within a given U.S. state, the average amount remitted does not vary systematically by the migrant's municipality of origin. While this assumption ignores micro-level heterogeneity, it provides a baseline estimate of the bilateral remittance flows that can be validated against the aggregate data provided by Banxico's series CE166.

3.3.1. Persistence of migration networks

The persistence assumption of migration networks is critical to our estimation strategy, as it justifies the use of MCAS data to construct the migration-based weighting matrix in order to allocate remittance flows. The stability of migration patterns over time is well-documented in the literature, with studies showing that established migration networks tend to dictate future migration patterns, creating persistent flows between specific origin and destination pairs (Card, 2001; Munshi, 2003).

To verify this assumption within our specific sample period, we conduct a validation exercise using the MCAS data by computing the Pearson correlation of our constructed weighting matrices across years. As shown in Figure 3.1, when taking any year from 2012 onward as the base year, the correlation with subsequent years is consistently above 0.90. While the 2010 and 2011 matrices exhibit strong correlation between them, their correlation with later years is smaller, specifically between 0.75 and 0.85. This may be due to structural adjustment in migration patterns following the Great Recession.

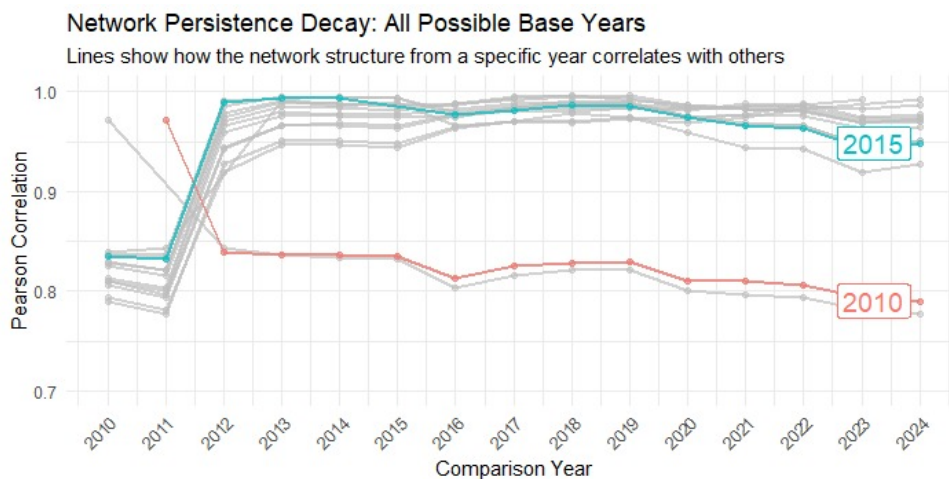


Figure 3.1: Correlation of MCAS-based weighting matrices across years. Each point represents the correlation coefficient between the weighting matrix of the base year and the weighting matrix of the comparison year.

Source: Authors' own elaboration based on MCAS data.

This high degree of correlation suggests that the spatial distribution of Mexican migrants across U.S. states has remained highly stable over time, lending credibility to our use of an averaged weighting matrix for estimating bilateral remittance flows.

3.3.2. *Validation of the estimated remittance flows*

Our methodology to estimate bilateral remittance flows implicitly assumes a homogeneous send behavior among migrants withing a given U.S. destination. That is, migrants in U.S. state s send on average the same amount of remittances, regardless of their specific Mexican municipality of origin m . To evaluate the empirical validity of this assumption, we aggregate our estimated bilateral municipal inflows and compare them against the observed municipal receipts reported in Banco de México's series CE166.

Figure 3.2 plots the Pearson correlation between our estimates aggregated at the municipal level and the official inflows for each year. The results report correlation coefficients consistently between 0.75 and 0.85, suggesting that our estimation procedure successfully captures a significant portion of the variation in remittance inflows across Mexican municipalities. While this result validates our baseline approach, we acknowledge that it does not capture all the heterogeneity in remittance flows, which is expected given the simplifying assumptions we made.

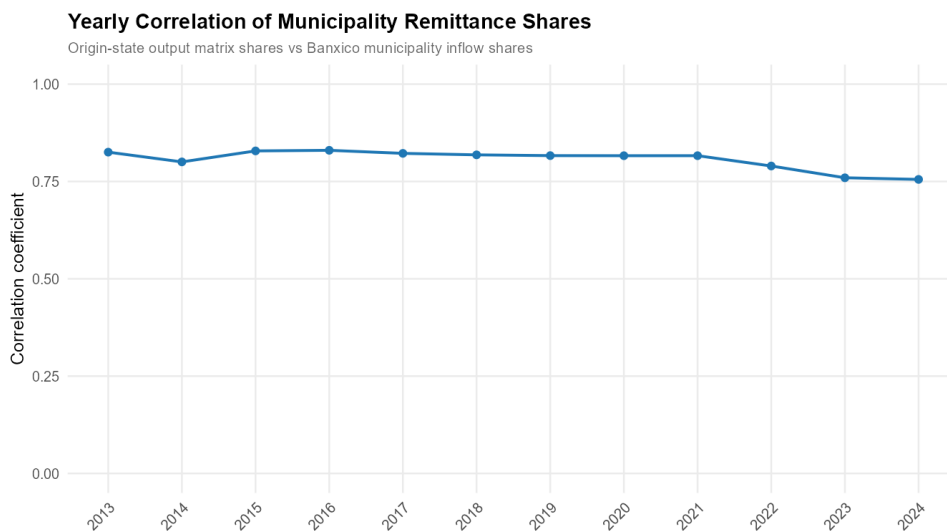


Figure 3.2: Correlation of estimated municipal remittance inflows with official data. Each point represents the Pearson correlation coefficient between the estimated municipal inflows (aggregated from our bilateral estimates) and the actual municipal inflows recorded in Banxico's series CE166 for a given year.

Source: Authors' own elaboration.

4. Results

4.1. Preliminary statistics

check parallel trends

statistics before (e.g. average over the past year), 1, 2, 3, and 4 quarters after the shock. I would do this with growth rates even if the estimation is in levels because quarterly growth rates are more interpretable and take into account the trend. Note that there is also seasonality to take into account.

4.2. DiD results

4.3. Arguing for Potential Spillover Effects

As argued above, given that the biggest hosts of Mexican migrants are Texas and California, it is likely that those municipalities exposed to the Florida shock might also be connected to a certain extent to California and Texas. Plot 4.1 helps us visualize that there exists indeed some migrant networks spanning across Florida and California and Florida and Texas.

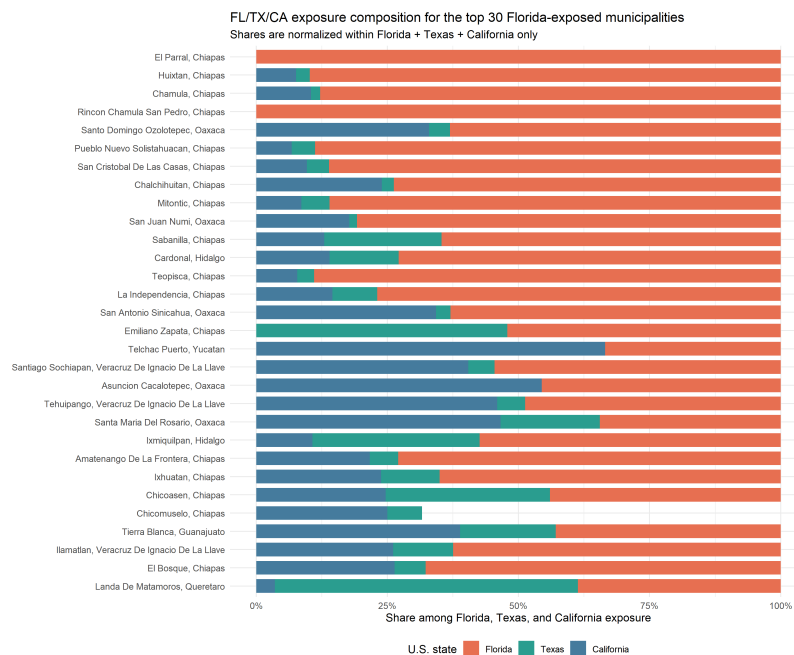


Figure 4.1: For the Top 30 Florida exposed municipalities, the plot shows the composition of migrants spread across the three US states.

Figure 4.2 suggests that there has been a dip in all 3 states in 2022Q4, after the Hurricane Ian hit Florida. However, we are interested in examining **how the inflow of remittances from Texas and California evolved around the shock relative to the evolution of Florida inflows**, specifically for the municipalities that are jointly highly exposed to Florida and California or Florida and Texas, provided there exist **parallel trends in the remittance patterns between these two groups (WE NEED PPML MAYBE!!!)**.

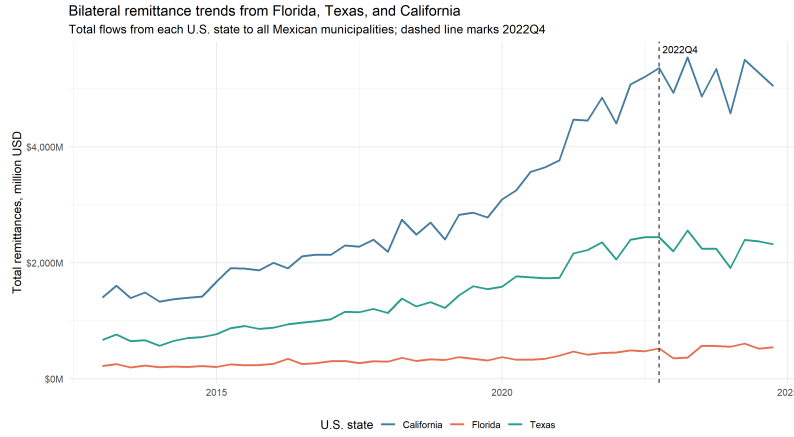


Figure 4.2: Evolution of Remittances Inflows from FL, TX and CA to all Mexican municipalities

Let's define a measure of joint exposure FL-TX of Mexican municipality m as the geometric mean of its exposure to Florida and its exposure to Texas, as follows:

$$JointExposure_m^{FL,TX} = \text{sqrt}(Exposure_m^{FL} \times Exposure_m^{TX})$$

Similarly, we can also define a measure of joint exposure FL-CA. As an example, 4.1 ranks the municipalities by highest joint exposure to both Florida and Texas.

We now define as "High joint exposure" as the top quartile municipalities with the highest joint exposure and "Low joint exposure" as the bottom quartile municipalities with the lowest joint exposure and estimate two log remittances regressions under the following specifications:

$$\log(1 + R_{mst}) = \alpha_t + \gamma_{ms} + \beta_{TX}(Texas_s \times Post_t) + \varepsilon_{mst}$$

$$\log(1 + R_{mst}) = \alpha_t + \gamma_{ms} + \beta_{CA}(California_s \times Post_t) + \varepsilon_{mst}$$

Note that in the TX-FL regression, the sample is restricted to the highly exposed municipalities (As defined above). Similarly, in the CA-FL regression, the sample is also restricted to the cor-

Table 4.1: Top 10 Joint Exposure FL-TX

Rank	State	Municipality	GeoMean	GeoMean %
1	Chiapas	Emiliano Zapata	0.3373	33.7261
2	Queretaro	Landa De Matamoros	0.3262	32.6221
3	Hidalgo	Pisaflores	0.3157	31.5685
4	Veracruz	Ignacio De La Llave	0.2703	27.0324
5	Michoacan De Ocampo	San Lucas	0.2673	26.7327
6	Chiapas	La Libertad	0.2539	25.3916
7	Veracruz	Tehuacan	0.2532	25.3234
8	Chiapas	Chicoasen	0.2391	23.9073
9	Chiapas	Sabanilla	0.2371	23.7081
10	Chiapas	Rayon	0.2348	23.4797

responding high jointly exposed municipalities.

We include α_t as the quarter FE, γ_{ms} as the corridor MX municipality - US State FE and $Post_t$ is defined such that $Post_t = 1$ for quarters from 2022Q4 onward, and 0 otherwise. Since Florida is the baseline category, the coefficient of interest β_{TX} (or β_{CA}) compares the post-2022Q4 change in Texas remittances to the post-2022Q4 change in Florida remittances (among our subsamples of interest). To ensure robustness of results, we also clustered at corridor and quarter level.

In other words, the coefficients β_{TX} and β_{CA} capture respectively:

$$\beta_{TX} = [\Delta \log R_{m,Texas}] - [\Delta \log R_{m,Florida}]$$

$$\beta_{CA} = [\Delta \log R_{m,California}] - [\Delta \log R_{m,Florida}]$$

The following figure 4.3 shows that in relative terms, the inflows of remittances from Texas and California to high joint exposure municipalities have experienced a significant increase in the two quarters posterior to the Hurricane shock, followed by an immediate decline. This could suggest a **short-term change of remittance patterns of these subsamples** right after the shock: municipalities that are strongly related to both Florida and the California or Texas might have been **relatively less affected by the Florida shock since they also received relatively more inflows of remittances from the other state that could potentially offset the decrease in inflows from FL**.

In other words, it could also suggest that the **REVISE: the estimates we found in the earlier sec-**

tions on the drop in remittances received by Florida households are actually attenuated by these potential spillover effects.

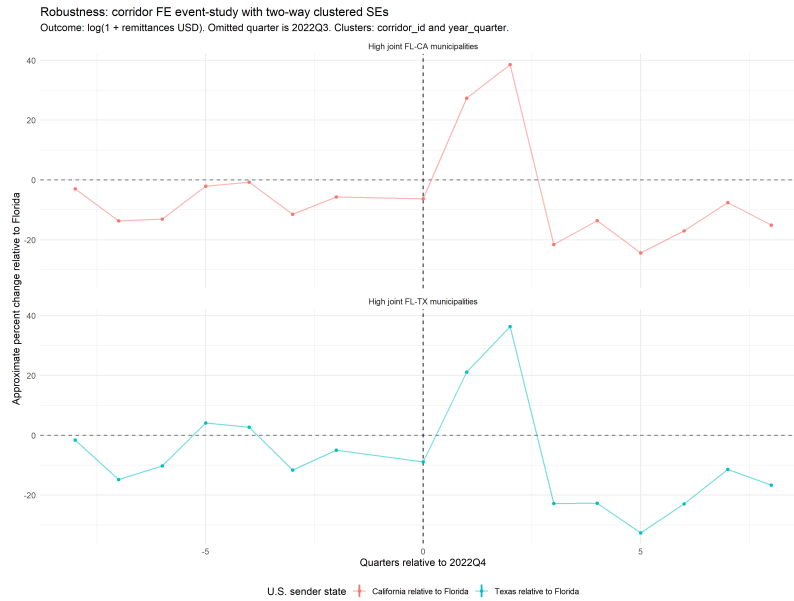


Figure 4.3: Change in log remittances inflows from TX and CA rel. to FL to high joint exposure municipalities

Despite the increase of remittances to high exposure municipalities from Texas and California relative to Florida, in order to support our intuition and confirm the existence of spillovers, we would need to check for parallel trends between the high and low exposure municipalities and their inflows of remittances from Texas/California. In other words, we do not want to **confound the compensating effect that has risen due to the Hurricane shock with a general upward trend in remittances received in Texas relative to Florida**, for instance.

here, we should actually test for parallel trends. control: low exposure municipalities TX-FL, treatment: high exposure municipalities TX-FL. We want to check if the trends in remittances received from Texas relative to Florida were parallel between the high and low exposure municipalities before the shock. If they were parallel, then we can be more confident that the divergence we see after the shock is indeed due to the shock and not due to some pre-existing trend.

5. Conclusion

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